Physics and Astronomy
University of Missouri Fall 2025

Communiqué

News From the Chair

Dear Alumni, Colleagues and Friends,

Greetings from the Department of Physics and Astronomy. As you will see from this newsletter, it has been a good year for our department.

Our program is as strong as ever, with 106 undergraduate majors and 63 graduate students enrolled in Fall 2025. The increased graduate enrollment, being above 60 for the last two years, is driven by increased grant funding and the recruiting of new faculty over the last three years. The department has developed many new course offerings, ranging from Astrobiology to Applied Machine Learning in Physics, and we have a new initiative to incorporate computer-based exploration throughout the curriculum. Introduction to Materials Science was offered for the first time as the inaugural course for the new Materials Science and Engineering Institute, in which the Physics and Astronomy department is a major partner.

The department established a new mentoring program where every undergraduate major meets with a faculty member each semester to promote their strong engagement with our program; it includes discussions about society memberships, research internships, physics careers and career paths. A new study room was opened this year where our undergraduate majors can study, socialize, and receive tutoring in any of their courses from graduate teaching assistants who are available in the afternoons. The department has many student organizations that are very active – the Physics and Astronomy Graduate Student Association (PAGSA), the Society of Physics Students (SPS), MU Student Astronomical Society (MUSAS), and Missouri Women in Physics and Astronomy (MoWiPA) – you can read about their activities in this newsletter.

Also in this newsletter you will see the many important research achievements of our faculty and students. Our faculty have received significant new research funding and external recognition for their research accomplishments. Additionally, two of our faculty were promoted this year. Dr. Deepak Singh was promoted to Professor. Dr. Silvia Bompadre, who serves both as the Associate Department Chair and the Director of Undergraduate Studies, was promoted to Teaching Professor.

Recipients of Graduate School and Graduate Professional Council Awards: Teja Teppala - the AAAS Case Workshop Award, Professor Suchi Guha - the Gold Chalk Award, Vicki Kuhn – the Excellence in Research Award.

After 40 years in the Physics Machine Shop, Rod Schlotzhauer retired in March. The shop has a long history of research instrument development for the department, the research reactor, the campus, and national research institutions. Rod was the shop supervisor, and we thank him for his excellent leadership.

This year we presented our inaugural Dorina Kosztin Memorial Endowment Award to undergraduate Julianna Bayless and graduate students, Dallar Babaian and Creighton Lisowski. The award was established in fond memory of Prof. Dorina Kosztin who was an outstanding and beloved faculty member for over twenty years. In the spirit of Dr. Kosztin's broad contributions to the department, the award recognizes the complete person for their accomplishments in all areas of research, teaching/mentoring, and service.

(cont. on page 5)

Inaugural Dorina Kosztin Memorial Endowment & Awards

Professor Dorina Kosztin

The Dorina Kosztin Memorial Endowment was established in early 2025 in fond memory of Curators Distinguished Teaching Professor Dorina Kosztin, who was an outstanding and beloved teacher in the Department of Physics and Astronomy. For over twenty years, Dr. Kosztin's wisdom, kindness, guidance, and mentoring, both in and out of the classroom, impacted countless students. It is this spirit of the complete person that guides the three Dorina Kosztin Memorial Awards (DKMAs), provided yearly by the endowment. The award categories are as follows:

1. The DKMA for Physics and Astronomy (P&A) Undergraduate Students: One or more annual scholarship awards shall be given to undergraduate students who are in the final year of their degree and excel in two of the three areas of research, teaching mentoring, and or service.

2. The DKMA for P&A Graduate Students: One or more annual scholarship awards shall be given to graduate students who are in the final year of their degree and excel in all three areas of research, teaching mentoring, and or service.

3. The Dorina Kosztin Memorial Service Award: One or more annual awards shall be given to outstanding faculty and or staff of the College of Arts and Science for outstanding and exemplary service to the College and or the University.

This inaugural year, recipients of the DKMAs were undergraduate student **Julianna Bayless**, and graduate students **Dallar Babaian** and **Creighton Lisowski**. They were selected by a Department Committee, nominated by the Chair. The DKMAs, consisting of a diploma, prize money, and a keepsake, were presented to the recipients on May 9, during the end-of-academic-year Department Luncheon. For all three recipients, the DKMA also has a sentimental value because they all knew Dr. Kosztin well and worked with her.

2025 DKMA recipients (from left to right): Dallar Babaian, Julianna Bayless, and Creighton Lisowski

New Alumni Endowments

Donald L. and Lona Lewis Packwood Endowed Assistant Professor of Applied Physics

Don Packwood with his late wife Leona in Pedernal, NW New Mexico in 2009

As alumnus and one of the co-founders of our Physics Leaders group, Don Packwood has generously contributed both his time and resources to our Department of Physics and Astronomy. His contributions are now able to endow a new position for assistant professor. The university has approved a search for this new position which will soon be underway. Don also funds an annual undergraduate student scholarship: the "Donald L. and Lona Lewis Packwood Endowed Undergraduate Scholarship Fund in Physics". Don received his PhD from our department in 1971 and followed that with a career at several companies, most of it at Hewlett Packard Corporation. He received an MU-A&S Distinguished Alumnus Award in 2004, and a Distinguished Alumni Award from our department in 2020.

Ronald J. Boain and Catherine J. Boain Endowed PhD Dissertation Award in Physics

and the

Ronald J. Boain and Catherine J. Boain Endowed Student Travel Fund

Catherine and Ronald Boain

Through the generosity of alumnus Ronald J. Boain the Department of Physics and Astronomy has been presenting an annual cash award for the top PhD dissertations and awarding funds to our students to enable them to attend conferences and other trave. These funds, originally set up in 2019, are now endowed in perpetuity and will benefit our students far into the future.

Additionally, Ron and Catherine established the Ronald J. Boain and Catherine J. Rangel Boain Endowment fund supporting travel to professional conferences and research sites for our graduate students.

Ron had a long and varied successful career in aerospace at TRW, Martin Marietta Aerospace, and the Jet Propulsion Laboratory. Ron was the 2024 recipient of the Department of Physics & Astronomy Distinguished Alumni Award and is one of our Physics Leaders.

Campus and Community Outreach

March Fair and Expo

Our department and students participated in the March 8, 2025, Columbia Young Scientist Fair and Expo, an annual outreach event organized by the Division of Research, Innovation, and Impact of the MU provost's office. The turnout of this event is usually 400-500. Our department has had a significant presence at this event for the past three years. This year we had four tables of displays and hands-on activities with a team of 10 student volunteers. The department also hosted an open house on April 25.

Physics & Astronomy Table

Laws Observatory

Laws Observatory on top of the Physics Building continues to be used for outreach and education on a weekly basis. Wednesday night viewing for the public is open (weather permitting) all year except during the coldest months and during a mid-summer break. There are many other activities for special events. The observatory often provides live streaming on Facebook at the observatory and off-campus locations. The observatory is run by student volunteers and by Valentine Germann and Randall Durk of the Central Missouri Astronomical Association.

Viewing sunspots at Logboat Brewery in August

July 9 photo from the roof

Val Germann was interviewed for the COMO Magazine Feb 12, 2025 issue regarding astronomy in Columbia: https://comomag.com/2025/02/12/reaching-for-the-stars-stargazing-in-columbia-mo/

For news and information follow Laws Observatory on Facebook and the observatory webpage: https://physics.missouri.edu/laws-observatory.

Wednesday nights at Laws Observatory

Science & Eng Collaborative Research at MU

Materials and Engineering Institute
MSEI, founded in 2021, is co-led by Matt Maschmann, professor of mechanical and aerospace engineering, and **Carlos Wexler**, professor of physics in the College of Arts and Science. The institute fosters collaboration in existing and new areas such as biomaterials, energy, materials, quantum materials and materials at extremes. MSEI's 2,000-square-foot Materials Characterization Facility is available for use and includes a variety of microfabrication and characterization devices including a Class 100 clean room. In addition to Prof. Wexler, our department has many professors included in the core faculty group: **Guang Bian**, **Suchi Guha**, **Tom Heitmann**, **Gavin King**, **Pontus Laurell**, **Andrew Meng**, **Paul Miceli**, **Sashi Sitpathy**, **Deepak Singh**, **Pavio Sukhachov**, and **Ping Yu**. Affiliated faculty include **Ioan Kosztin** and **Carsten Ulrich**.
Source: *https://msei.missouri.edu*

Quantum Innovation Center (Mizzou-QIC)
Physics Asst. Professor **Pontus Laurell** is a member of the executive committee of the recently launched Quantum Innovation Center (Mizzou-QIC) connecting faculty, students, and industry partners interested in quantum computing. The center is a dynamic hub for quantum research, education, and innovation. In addition to Laurell, affiliated department faculty include **Shi-Jie Chen**, **Carlos Wexler**, and **Xiaoqin Zou**. The QIC accepts student research proposals for its summer internship program. Several students were funded for this summer's inaugural program, including **Amarnath Chakraborty**, one of our physics graduate students. Source: *https://engineering.missouri.edu/quantum-innovation-center*

*(**News From the Chair**, cont. from page 1)*

The department is extremely grateful for several new endowments that were established this year – these will have an enormous impact on the Physics and Astronomy program. The Donald L. and Lona Lewis Packwood Endowed Assistant Professor of Applied Physics will support a new junior faculty member whose focus will be on physics applications. A faculty search commenced this fall with the intent to hire for Fall 2026.The Ronald J. Boain and Catherine J. Boain Endowed Student Travel Fund will assist our students in travelling to professional conferences and research sites.

The department has an enormous appreciation of the generosity of our alumni who have contributed to scholarships and fellowships that play an essential role in supporting our students. This year the department awarded the John H. Letcher Graduate Student Scholarship for the first time and next year the Chandrasekhar Fellow will be presented to an outstanding graduate doctoral student in their last year. We are also grateful for our Physics Leaders alumni who contribute their time and energy to the department. Our annual Physics Leaders meeting was held October 3-4, 2025, where they engaged with the faculty and students. The department presented the Distinguished Alumni Award to Dr. Paul Leath (PhD 1966) for his distinguished career and long-time service to the department. The Physics Leaders will meet again during one of the first three weekends in October 2026 – all alumni are welcome, so please watch your email for a final date and I hope to see you next year!

Wishing everyone the best for the new year, Paul F. Miceli

New Chandrasekhar Fellow

Through the generosity of Meera and H. R. Chandrasekhar, the Department of Physics and Astronomy is gaining a new one-semester fellowship (.5 FTE) with an inaugural presentation next spring semester. The awardee will be an outstanding graduate student in their final year, who plans to defend their PhD dissertation by May 2026. The outstanding student will be designated as a "Chandrasekhar Fellow". The department appreciates this generous support by Meera and Chandra, both professors emeriti of our department, in endowing this fellowship and for their years of dedication to research, teaching, and mentoring.

In the Future: Next Gen MURR

The University of Missouri plans to build a NextGen MURR (Next Generation Missouri University Research Reactor) over the next several years, a multi-billion-dollar initiative to accelerate the production of cancer-fighting radioisotopes. When completed it will be the most powerful university research reactor in the U.S. The new 20-megawatt research reactor along with our existing reactor will serve as a hub for nuclear medicine innovation. Currently our 10MW reactor annually supplies 1.6 million patients with radioisotopes that treat or diagnose cancer. The new next-generation facility will be located at Discovery Ridge Research Park at Highway 63 and Discovery Parkway in south Columbia.

Using state and federal funds, MU signed an agreement with the Korean Atomic Energy Research Institute (KAERI) in March for the initial design and licensing phase. News regarding the signing can be found at: https://www.ans.org/news/2025-04-16/article-6939/mizzou-announces-contract-signing-for-nextgen-murr-project/

Importantly, there is a proposal for a National Neutron Scattering User Facility adjacent to Next-Gen MURR and KAERI's preconceptual design will include a Cold Source, Guide Tubes, and Guide Hall. This facility would serve a significant national need for more capacity in neutron scattering and it would be a transformative addition to our campus.

Sources: KOMU.com. Link to NextGen MURR: https://nextgenmurr.missouri.edu/

A Stellar Hobby

Bangzheng (Tom) Sun, an astrophysics graduate student working with Haojing Yan, does more than pursue research in astrophysics. He is also a skilled astrophotographer capturing high resolution images of deep sky objects.

Tom combines hours of exposure time at various locations, including Finger Lakes State Park (Bortle of 5), extremely dark skies in New Mexico (Bordle of 1), and even our building roof (Bortle of 7) to create these amazing deep sky images

Stephens Quintet, NGC 7331, & Supernova SN2025. Total observing time 7.17 hours. Photo by Bangzheng (Tom) Sun

Pillars of Creation (Messier 16). Total observing time: 17.42 hours. Photo by Bangzheng (Tom) Sun

Tom states that his telescope is also JWST: "Just a Weak & Small Telescope", but his results are certainly impressive. The picture on the lower left corner shows this "JWST" running an imaging session with the beautiful constellation of Orion rising on the background.

The Bortle Scale ranges from Class 1 (darkest sky available) to Class 9 (inner city skies)

The Shark Nebula. Photo by Bangzheng (Tom) Sun

"JWST" running an imaging session for the constellation Orion

Faculty News & Recognition

Silvia Bompadre was promoted to **Teaching Professor**. Prof. Bompadre is also the Associate Department Chair and Director of Undergraduate Studies.

Curators' Distinguished Professor Shi-Jie Chen received a grant for $2,609,232 from the NIH to study "New methods for computational modeling of RNA structures". Prof. Shi-Jie Chen's lab was ranked #1 in the CASP competition for RNA structure prediction. He is also cited and quoted in a recent Nature (March 24, 2025) about the competition and the new focus on RNA structures.

Asst. Professor Keith Cassidy was one of several MU researchers studying the structure of a protein that could lead to more targeted heart disease treatments. Their work "A Protein at the heart of heart disease" was highlighted in Show Me Mizzou: *https://showme.missouri.edu/2025/a-protein-at-the-heart-of-heart-disease/*.
Researchers Zachary Berndsen and Keith Cassidy were recruited to Mizzou because of their knowledge of cryo-electron microscopy, a technique that uses electrons to determine the 3D structure of biological molecules. Their recent paper, "The structure of apolipoprotein B100 from human low-density lipoprotein"
was published in Nature:
https://www.nature.com/articles/s41586-024-08467-w.
Their article "The structure of bad cholesterol comes into focus" was also published in Nature:
https://www.nature.com/articles/d41586-024-04210-7

Professor Suchi Guha was awarded the Gold Chalk Award in 2025 by the Graduate Professional Council for significant contribution to graduate education. She and her research group have been very active working in areas of neuromorphic organic devices and ultrafast optical spectroscopy. She gave several invited international presentations including the Nanoseries Conference in Valencia, Spain, Electronic Structure Theory and Experimental Realization in IIT Madras, and the International Materials Research Congress in Cancun. Graduate student, Dallar Babaian (co-advised by Prof. Ping Yu), defended her PhD dissertation in August 2025 and has recently joined Lawrence Berkeley National Laboratory as a postdoctoral scholar.

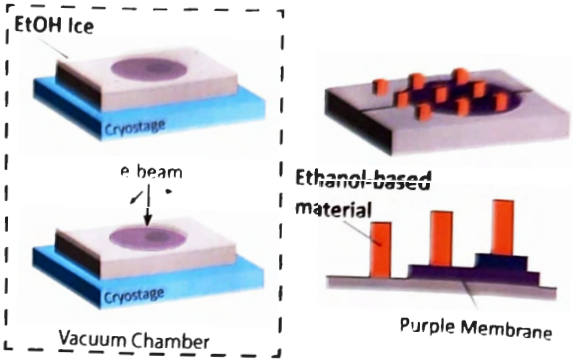
Suchi Guha with students Randy Burns and Dallar Babaian

Randy Burns was awarded the 2024 Ron Boain and Catherine Rangel-Boain Dissertation Award. Arash Ghobadi presented his work on neuromorphic organic transistors at the 2025 spring meeting of the MRS (Materials Research Society). Arash, Dallar, Randy, and undergraduate students Evan Restuccia and Thomas Kallaos have authored/co-authored several publications.

Professors Suchi Guha and **Gavin King** were featured in a Show Me Mizzou article on their collaborative research harnessing perovskites) for energy innovation. *https://showme.missouri.edu/2025/mizzou-scientists-harness-the-power-of-layered-crystals-for-energy-innovation/*

Professor Gavin King's laboratory had an active year. They welcomed a new graduate student (Alankit Dey) and several undergraduate researchers (Kira Wilson, Oktavia Jaszczynska).

The group published several high-profile papers including, most recently, in Nano Letters. In this NSF-funded work they employed their newly refurbished Ice Lithography instrument and developed a novel use of frozen ethanol vapor for electron beam lithography on biological material. In particular, they demonstrated the fabrication of solid graphite-like material directly on biological membranes, portending future applications such as living solar cells. A patent disclosure has been filed to facilitate future commercialization of the technology.



The above cartoon shows the concept of using frozen ethanol (EtOH ice) as a sacrificial resist in an electron beam (e-beam) lithography process. The e-beam converts the frozen ethanol into a graphite-like material (ethanol based material, orange) that is solid at room temperature. The strategy enables precise 3D printing on delicate biological membranes. Cartoon source: from D. Chiaro *et al.*, Nano Letters 2025.

Asst. Professor Pontus Laurell serves on the executive committee of MU's Quantum Innovation Center (Mizzou-QIC) connecting faculty, students, and industry partners interested in quantum computing. The QIC, through its Internship program, funded several student projects over the summer, including Amarnath Chakraborty in physics.

Prof. Laurell gave an invited talk at the SCES2025 (Strongly Correlated Electron Systems) conference in Montreal, that was part of a special symposium session on Entanglement in SCES which was supported by a Faculty International Travel Award from the College of Arts and Science.

Asst. Professor Maria Mills was the recipient of a National Institutes of Health (NIH) award funded at $350,000 annually for 5 years, a total of $1.7 million, for her research on the coordination of complex dynamics by DNA topoisomerases and DNA repair processes. Prof. Mills' group uses a combination of force, torque, and fluorescence to understand how proteins remodel the structure of DNA.

Deepak Singh was promoted to **Professor.** Prof. Singh's research along with graduate student Pousali Ghosh on their important discovery of magnetism in NiSi is featured on the Department of Energy – Basic Energy Sciences website:
https://www.energy.gov/science/bes/articles/surprising-discovery-magnetism-common-material-microelectronics
and also featured in Show Me Mizzou last fall:
https://showme.missouri.edu/2024/revolutionizing-electronics/

Singh (right) works with Mizzou graduate student Pousali Ghosh in Singh's lab.

Asst. Professor Charles Steinhardt's recent research offering a new theory on star formation was recognized in a Show-Me Mizzou article in April:
https://showme.missouri.edu/2025/mizzou-researcher-offers-new-theory-on-universes-star-formation/

Curators' Distinguished Professor Carsten Ullrich has received a Fulbright Specialist Program Award and will be on the Fulbright Specialist roster for three years (2024-2027). He will spend October 2025 at the Indian Institute of Technology (IIT) Madras in Chennai, where he will teach a graduate course on Introduction to Time-Dependent Density-Functional Theory, and will pursue a research collaboration with scientists at the Condensed Matter Theory and Computational Lab (CMTCL).

Professor Haojing Yan and his team have been working on a few projects in extragalactic astronomy that are leading to some interesting (and unexpected) results. In particular, he and one of his graduate students, Bangzheng (Tom) Sun, have published two significant papers using the data from the James Webb Space Telescope (JWST). One is on the early settlement of Milky Way-like spiral galaxies, and the other is on the discovery of candidate galaxies in the early universe that are unusually luminous. The latter has been publicized by Show Me Mizzou in August 2025 and received world-wide attention - it is in the top 5% of all research outputs scored by Altmetric. They have three more papers currently under review.
https://showme.missouri.edu/2025/early-galaxies-or-something-else-mizzou-scientists-uncover-mysterious-objects-in-the-universe/

Our Graduates

Undergraduate Degrees

BS, Fall 2024

Dylan Bateman, *emphasis in Astronomy*
Matt Gardner, *emphasis in Astronomy, Cum Laude*
Brian Issleib
Ash Morrison, *emphasis in Astronomy, Magna Cum Laude, Honors College Certificate*
Kyle Renshaw, *Summa Cum Laude*
Evan Restuccia, *Summa Cum Laude*
Alec Williams, *emphasis in Astronomy*

BS, Spring 2025

Julianna Bayless, *emphasis in Astronomy, Honors College Certificate, Department Honors*
Emma Burton, *emphasis in Astronomy*
Mason Crane
Carter Davidson, *emphasis in Materials Science*
Patricia Denn, *emphasis in Materials Science*
Donovan Flagg, *emphasis in Astronomy*
Ellen Gates, *Emphasis in Astronomy, Magna Cum Laude, Honors College Certificate, Department Honors*
Cy Jennings, *Emphasis in Astronomy*
Carter Woods, *Magna Cum Laude*

BS, Summer 2025

Vishvi Aurora

Departmental Honors requires > 3.5 GPA in physics courses, completion of 6 credit hours of research, and a publication or presentation (oral or poster).

A Certificate in General Honors from the MU Honors College requires students to complete 24 hours of courses for honors credit and maintain a 3.5 cumulative GPA.

To receive Latin honors from our College of Arts and Science requires at least 54 of the student's last 60 hours at MU and:

3.7 – 3.799 GPA for Cum Laude
3.8 – 3.899 GPA for Magna Cum Laude
3.9 – 4.0 GPA for Summa Cum Laude

Graduate Degrees

PhD Graduates:

Fall 2024

Jared Williams, Advisor: Carsten Ullrich

Spring 2025

Charles Mentzer, Advisor: Aigen Li
Alec Martin, Advisor: Yicheng Guo
Vishal Jayswal, Advisor Sergei Kopeikin

Summer 2025

Jacob Cook, Advisor Guang Bian
Ehsan Faridi, Advisor Giovanni Vignale
Teja Teppala, Advisor Yicheng Guo
Already defended, Fall 2025:
Dallar Babaian, Advisors Suchi Guha and Ping Yu
Pousali Ghosh, Advisor Deepak Singh
Amarnath Chakraborty, Advisor Giovanni Vignale

Spring 2025 PhD Graduates

Professor Sergei Kopeikin, Vishal Jayswal, Charles Mentzer, Alec Martin, Professor Yicheng Guo

Where are our graduates?

Dallar Babian
PhD, August 2025
Advisor: Suchi Guha
Postdoc, Lawrence Berkeley National Lab

Jacob Cook
PhD, summer 2025
Advisor: Guang Bian
Postdoc, Oakridge National Lab

Ehsan Faridi
PhD, summer 2025
Advisor: Giovanni Vignale
Lecturer, Savannah State U.

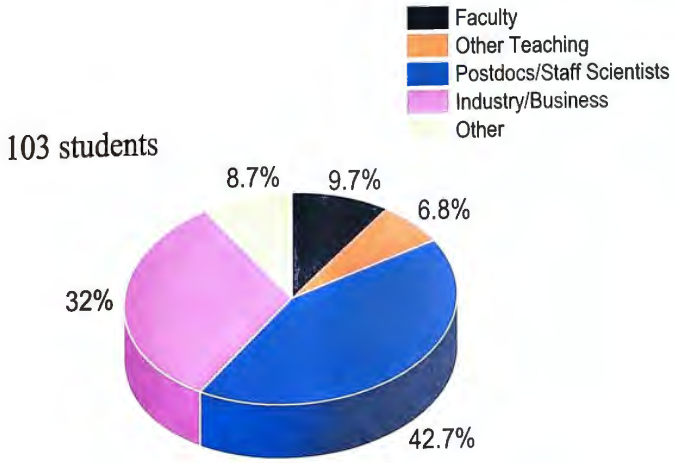
Pousali Ghosh
PhD, August 2025
Advisor: Deepak Singh
Postdoc, Howard University

Vishal Jayswal
PhD, spring 2025
Advisor: Sergei Kopeikin
Instructor AP Physics, Lawrence HS, KS

Alec Martin
PhD, spring 2025
Advisor: Yicheng Guo
Lecturer, Troy University

Charles Mentzer
PhD, spring 2025
Advisor: Aigen Li
Postdoctoral fellow (National Tsing Hua Univ)

Jared Williams
PhD, fall 2025
Advisor: Carsten Ullrich
Lecturer, Stephens College



PhD Physics, U. Missouri (2010 - 2025)

Student Awards, Scholarships, & Recognition

Undergraduate Student Scholarships

Paul E. Basye Undergraduate Scholarship
Julianna Bayless, Mathieux Harper, Ashley Morrison

Samuel S. Laws Scholarship
Mathieux Harper

Donald L. and Lona Lewis Packwood Endowed Undergraduate Scholarship Fund in Physics
Juliana Bayless

Melvin Y. Mora Undergraduate Scholarship Fund
Ashley Morrison

Clifford W. Tompson Scholarship in Physics
Vishvi Aurora, Carter Woods

These fellowships, scholarships, and awards are possible because of funds generously established and supported by our alumni and faculty.

These students were the recipients for the academic year 2024-2025 and summer 2025.

Some funds do not have scholarships awarded every year.

Graduate Student Scholarships

Rose Marie Dishman Endowed Scholarship in Physics
Tapesh Gautam, Arash Ghobadi, Gourab Nandi

James L. and Dora D. Fergason Fund for Excellence in Physics:
Ragheer Yeassin, Kristopher Luck, Briana Montalvo, Ashan Gunarathna

Newell S. Gingrich Physics Scholarship Fund
Dilan Gamachchige, Indeewari Kariyapperuma, Bangzheng (Tom) Sun

Newell S. Gingrich Physics and Astronomy Endowment
Dayton Simpkins, Kishlay Singh

Eli Stuart Haynes and Nola Anderson Haynes Scholarship Fund
Harold Diaz-Quiroz

Ernest W. Landen Fellowship in Physics
Shakeila Davis, Pavel Morales

B. H. Rose Endowment Fund
Bipin Bhattarai

Guy Schupp Scholarship Fund
Connor Nance

Armstrong Physics Endowment
Stephen Klue, Gourab Nandi

Letcher Graduate Student Scholarship
Jenna Bologa, Dominic Dalba, Bradley Faison

O. M. Stewart Scholarship
Abhi Abhijeet, Bibek Tiwari, Abeeb Ajibade, Mohammed Abousetit, Ishan Pathak, Dallar Babaian, Dillon Balthrop, David Beckwitt, Moudip Nandi, Deepesh Sigdel, Amarnath Chakraborty, Ehsan Faridi, Pousali Ghosh, Teja Teppala, Mari Tsumuraya, Bo Yang, Mitch Vaninger, Fatemah Zohra, Akansha Negi, Prativa Sharma, Erin Brady, Geroge Doering, Aparna Joshi, Diya Bansal, John Horne

Department Graduate Student Awards

Ron Boain and Catherine Rangel-Boain Dissertation Awards (fall 2024)
Randy Burns (first place), Ryan Smith (second place)

Dorina Kosztin Memorial Award
Dallar Babaian, Creighton Lisowski
Julianna Bayless (undergrad)

Harry E. Hammond Teaching Assistant Award
Mohamed Abousetit, Marek Spader, Stephen Klue

Ron Boain and Catherine Rangel-Boain Travel Award
Abeeb Ajibade, Amarnath Chakraborty, Anthony Dorhauer, Arash Ghobadi, Avinash Sah, Clayton Connor

Carl and Brynn Anderson Travel Award
Creighton Lisowski, Dallar Babian, Deepesh Sigdel, Dillon Balthrop, Jacob Cook, Jenna Bologa, Sadie Nickles, Stephen Klue, Tapesh Gautam, Mari Tsumuraya, Pousali Ghosh

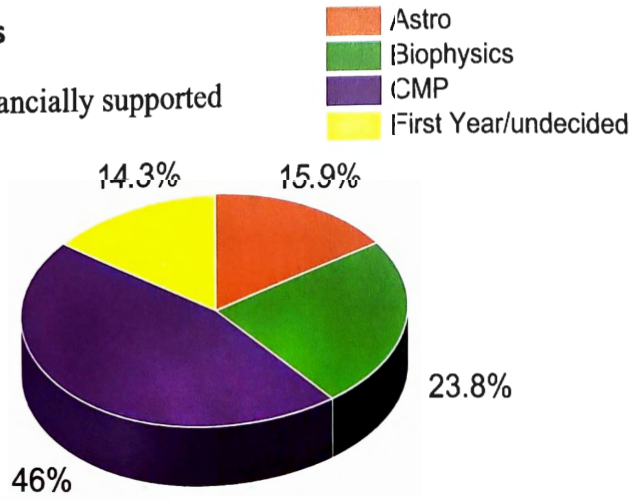
Spotlight on our Scholarship Funds

This is the first year that student awards were given from the Letcher Graduate Student Scholarship Fund. John H. Letcher III. graduated with a PhD in Physics from our department in 1963, He worked for McDonald Douglas, Monsanto, and other companies in industry, and was later was a professor of computer science at the University of Tulsa. Dr. Letcher passed away in 2023. We appreciate that he generously included our department in his legacy planning, benefitting our students far into the future.

Overall size of Physics and Astronomy graduate program in Fall 2025:

63 students

All students are fully financially supported



External Student Awards

Vicki Kuhn receives award from Jessica Osaze, past president of GPC

Graduate student **Vicki Kuhn** received the 2025 MU Graduate Professional Council Excellence in Research Award. This award honors a graduate or professional student who has demonstrated a sound understanding of their field of study and has made contributions to the field through their research. Vicki's advisor is Yicheng Guo.

Graduate student **Gourab Nandi** received the American Astronomical Society FAMOUS (Funds for Astronomical Meetings: Outreach to Underrepresented Scientists) Travel Grant which supported his travel to 246th American Astronomical Society Meeting in Anchorage, Alaska. He also received the Chambliss Astronomy Achievement Student Award-Honorable Mention, for his poster presentation at the meeting recognizing exceptional student research.

Graduate student **Amarnath Chakraborty** was accepted with the Quantum Innovation Center by IBM (Mizzou-QIC) for its competitive inaugural summer internship program. Mizzou-QIC is a hub for quantum research, education, and innovation at MU and across the MU System. Amarnath was accepted to present at a Quantum Developer Conference hosted by IBM. Amarnath's advisor is Givanni Vignale.

Graduate Student **Teja Teppala** was selected in March as one of the recipients of the Mizzou 18 award. This award is dedicated to honoring 18 University of Missouri graduate and professional students in the last year of their degree eligibility, chosen for their research, collaboration with faculty and staff, and their demonstrated leadership with undergraduate students. Teja was also selected to represent Mizzou Graduate School at the 2025 Catalyzing Advocacy in Science and Engineering (CASE) Workshop held by the American Association for the Advancement of Science (AAAS) in April. Teja's advisor is Yicheng Guo.

Senior physics major **Donavan Flagg** receive an award at the Show Me Research week April 2025 for "Measuring the pitch angle of distant galaxies". Donovan is mentored by Yicheng Guo.

Senior physics major **Ellie Gates** received the University of Missouri Award for Academic Distinction in 2025. This award recognizes extraordinary intellectual curiosity, seeking knowledge beyond the classroom, sharing that knowledge with public audiences, and contributing to the academic atmosphere at the university. Ellie works with Yicheng Guo.

Our New Graduate Students

Matthew Bruenning
(Missouri State U)

Xinjlun Chen
(UC Stanta Cruz)

Hyrum Harlow
(Missouri State U.)

Lichun Ji
(Columbia U., NY)

RongHao Luo
(Southern U of Science & Technology, China)

Griffin Nettelhorst
(U. of Illinois, Urbana)

Jakob Perivolotis
(Arizona State U.)

Fahmida Rahman
(U. Barishal, Bangladesh)

Leonardo Ramos
(Lewis U., IL)

Henri Reno
(Knox College, IL)

Student Organizations

PAGSA

This year, the Physics and Astronomy Graduate Student Association (PAGSA) has continued to bring students together through a variety of fun and community-building events. In February, members gathered for a cozy movie night featuring *The Imitation Game*, a film that resonated with both the scientific and social sides of graduate life. March was dedicated to service, as the group organized a library cleanup and reorganization project, helping maintain a welcoming and useful space for everyone in the department. Spring wrapped up with an exciting trivia night in April and a relaxing afternoon of disc golf followed by a department-wide picnic in May, giving students a chance to enjoy the outdoors and some friendly competition.

The fall semester kicked off in August with a welcome picnic to greet new and returning students, an event that set a warm and inclusive tone for the year. In September, PAGSA hosted a game night where members could unwind, share laughs, and build connections outside of the lab and classroom. Alongside these monthly activities, the group also maintains regular traditions such as the monthly association meeting, where graduate students discuss upcoming plans and share ideas, and TAP Hour, a recurring event where students give short talks on their research or topics of interest. Together, these events foster both academic engagement and community among physics and astronomy graduate students.

Handmade piñata at spring picnic

News contributed by Dayton Simpkins, PAGSA president

Spring picnic

Library cleanup

Scan this QR code to Follow PAGSA on Facebook and stay up to date.

Mizzou Women in Physics and Astronomy

MOWiPA (Mizzou Women in Physics and Astronomy) enjoyed another active year! We continue in our aim to represent, support, and give voice to the women and gender minorities of the Mizzou Physics and Astronomy department.

Fall pumpkin painting

Our bimonthly meetings include various social and cultural events to help build community between our undergraduate students, graduate students, and faculty. We work to facilitate the academic pursuits of our members, and to provide a forum where concerns can be raised, and guidance can be sought after.

In the fall, we had many festive events, such as pumpkin painting, movie nights, and gingerbread house decorating. In the spring, we explored topics such as the history of women in astronomy as we practiced our presentation skills. We also had a handful of members attend the Conference for Undergraduate Women and Gender Minorities (CU*iP)!

This year we hope to continue our traditions and grow our membership!

Department women at the Leaders Meeting dinner in October

News contributed by MOWiPA president Brianna Montalvo.

MU Student Astronomical Society

The Student Astronomical Society (SAS) is open to any Mizzou student interested in expanding their knowledge about the universe and all it encompasses. The SAS meets the second Wednesday of each month. They have monthly speakers and often work with the Central Missouri Astronomical Association for astronomy educational events.

Society of Physics Students

The Society of Physics Students (SPS) had a busy year. After wrapping up the summer with homemade liquid nitrogen ice cream, we hosted a series of student-led presentations on current physics topics. In November, we co-organized a Physics and Astronomy Scholars Conference with Mizzou Women in Physics and Astronomy (MOWiPA). Attendees had the opportunity to present research, learn about graduate school, and participate in programming workshops.

Chicago trip, Millenium Park

Over winter break, SPS traveled to Chicago to visit Fermilab. We had the opportunity to learn about high energy particle physics and the discovery of elementary particles like the tau neutrino. We also visited the Field Museum, the Shedd Aquarium, and the ice rink in Millenium Park.

During the Spring semester, SPS continued with research presentations, physics trivia, and more. At our largest meeting of the year, we worked together to build a cloud chamber.
Last Thursday, SPS held its first meeting of the new semester. We plan to build a new and improved cloud chamber, along with other projects in the coming year. We will visit another national laboratory over winter break.

Chicago Trip

News contributed by SPS President Harrison Davison.

Physics Leaders Fall Meeting

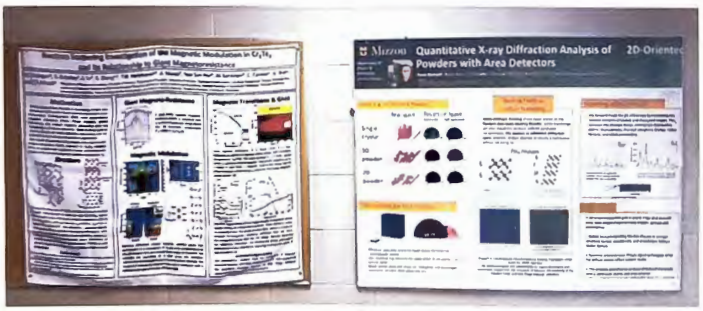
Our Physics Leader alumni meetings are held annually. The Physics Leaders are always interested in adding new members, and any alumnus can attend the event.

Our fall meetings always include briefings from Department Chair, Prof. Paul Miceli, Director of Graduate Students Prof. Suchi Guha, and Director of Undergraduate Studies Prof. Silvia Bompadre. Our students provide updates on our student organizations. The Dean of A&S Cooper Drury drops by to talk about A&S topics and answer questions from the alumni. The department selects up to four undergraduate and four graduate students to make 15-minute research presentations. As part of their giving back to the department, Physics Leaders evaluate and award monetary prizes to selected student presenters. A Friday evening event with socializing and dinner is held where these student awards, faculty recognition awards, and a department Distinguished Alumni are announced and awarded.

Paul Leath

This year the annual meeting was held on October 3-4, 2025. During the Friday morning meeting our department Distinguished Alumni Award was presented via Zoom to Paul Leath, PhD 1966, for his career achievements and longtime service to our department, including as one of the founders of the Physics Leaders group. His career in academia in condensed matter physics at Rutgers University included serving as vice-provost, provost, and later as Chair of the Physics Department.

This year we had three of our research labs plus a rooftop solarscope available for tours. The labs included semi-conductor crystal growth, ice lithography and an ultra-fast laser. Posters that included astrophotography, x-ray diffraction, neutron scattering, and ice lithography were on display on the second floor.



Posters on display

Graduate student Stephen Klue explains the ultra-fast laser setup to leaders Philip Chumbley and Chuck Crowder

Physics Leaders Ron Boain, Don Packwood, and Jim Seeser on the roof for solar viewing (sunspots)

The Friday evening dinner event was held at the Holiday Inn executive Center. Department chair Paul Miceli presented faculty recognition awards to Sergei Kopeikin, Guang Bian, Wouter Montfrooij, and Maria Mills.

Guang Bian and Sergei Kopekin received the faculty awards presented by Paul Miceli on behalf of the Physics Leaders.

Zoom presentation of Boain Dissertation Awards, Jacob Cook pictured.

The Boain Dissertation awards were presented by Paul Miceli and Ron Boain via Zoom to Jacob Cook (first place, advisor Guang Bian), and Alec Martin (second place, advisor Yicheng Guo).

Vicki Kuhn and David Beckwitt receive first and second place awards, respectively, from Vann Priest and Paul Miceli

Paul Miceli and Leaders President Vann Priest presented awards to the students the leaders selected as the top presenters on Friday afternoon, a difficult job to choose among many outstanding presentations.

- Undergraduate awardees: First place Carter Meyerhoff, second place Zach Borowiak
- Graduate awardees: first place Vicki Kuhn, second place tie to Sadie Nickles and David Beckwitt.

Saturday morning included a productive and extended leaders' discussion with Silvia Bompadre, Director of Undergraduate Students, and Suchi Guha, Director of Graduate Students.

Next year's meeting dates will be finalized later but will be one of the first three weekends in October.

Following the meeting the leaders, represented by leader's president Vann Priest composed a letter to the A&S Dean. The leaders commented on a stable department and one that is well-positioned to grow. One comment was: "The university's vision for science and engineering (Center for Energy Innovation, Material Science and Engineering Center, NextGen MURR and the National Neutron Scattering User Facility) will provide ample opportunities for the department."

Physics leaders Ron Boain, Don Packwood, and Jim Seeser on the roof of the physics building for solar viewing

Department Retirement

Rod Schlotzhauer retired in February after 40 incredible years of service in the Physics Machine Shop. The department held a farewell luncheon on February 26 to thank Rod for everything he has done for MU Physics & Astronomy, Mizzou, and countless external customers all over the world. The Physics Machine Shop is a leader in the design and construction of instrumentation for physics research and has attained a worldwide reputation for the development of instrumentation for neutron scattering.

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Update your contact information, through the MU Alumni Association website: website: http://www.mizzou.com/.
You do not need to be a member to update your information.

Do you know a department alumnus that you want to nominate for a Department of Physics & Astronomy Distinguished Alumni Award? Visit this website for more information:
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